Computational Physics

Summer 2026

Lecture #16

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Ordinary Differential Equations (ODEs)

- describe dynamics of many physical systems
  - ODE: derivatives w.r.t. only one variable
  - PDE: partial derivatives of several variables

$$\frac{dy}{dt} = ky \qquad \text{exponential growth} \qquad [t = \text{independent variable}]$$

$$\frac{\partial y}{\partial t} + c\frac{\partial y}{\partial x} = 0 \qquad \text{transport equation}$$

<u>Def</u>: an ODE is <u>order n</u> if its highest order derivative is n^{th}-order, i.e. $y^{(n)}$ or $\frac{d^n y}{dt^n}$

↳ in general, write $\boxed{y'(t) = \frac{dy}{dt} = f(t,y)}$ is a 1^{st} order ODE

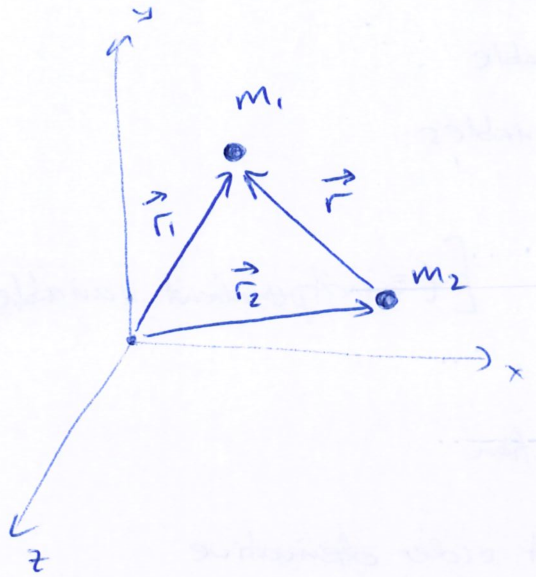
and with $y(t_0) = y_0$ specifies a general initial-value boundary (IVB) problem

<u>two notes</u>:

(1) can have $y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ i.e. several equations specifying a <u>system of differential equations</u> for y_i

(2) any order n ODE can be reduced to a first order system of ODEs

Ex: gravitational 2-body problem



- consider two point masses separated by a distance vector

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

with length and direction

$$r = |\vec{r}| \qquad \hat{r} = \frac{\vec{r}}{|\vec{r}|}$$

then Newton's equation for the gravitational acceleration:

$$\ddot{\vec{r}} = \frac{d^2\vec{r}}{dt^2} = -\frac{GM}{r^2}\hat{r}$$

where $M = m_1 + m_2$ is the total mass

$G = 6.67 \times 10^{-8}\ cm^3\, g^{-1}\, s^{-2}$ is the grav. constant

[this is a 2^{nd} order ODE]

let $\vec{F}_{12}$ be the force on m_1 exerted by m_2

$\vec{F}_{12} = -\vec{F}_{21}$ equal and opposite

$$\Rightarrow m_1\vec{a}_1 = -m_2\vec{a}_2 \quad \Rightarrow \quad \vec{a}_2 = \frac{-m_1}{m_2}\vec{a}_1$$

consider the relative acceleration between the two bodies:

$$\vec{a} = \vec{a}_1 - \vec{a}_2 = \vec{a}_1 + \frac{m_1}{m_2}\vec{a}_1$$

$$= \frac{m_1 + m_2}{m_1 m_2}\vec{a}_1 m_1$$

is $\vec{F}_{12}$

$\Rightarrow$ define the "reduced mass" $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$

rewriting, we have:

$$\vec{a} = \frac{1}{\mu} \vec{F}_{12}$$

$\Rightarrow \mu \vec{a} = \vec{F}_{12}$ i.e. m_1 moves with respect to m_2 like a single body with reduced mass μ

note: $\mu \leq m_1$ and $\mu \leq m_2$

and: $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}$ "reciprocal additive property"

$\lim_{m_1 \gg m_2} \mu \simeq m_2$ i.e. m_1 = Sun, m_2 = Earth

[m_1 is essentially stationary, and the relative motion of m_2 looks like that of a single mass with m_2] w.r.t. the center of mass

$\Rightarrow \mu \vec{a} = -\frac{G m_1 m_2}{r^2} \hat{r}$ the mutual acceleration

is a second order ODE, we can rewrite as a coupled system of two first order ODEs:

$$\boxed{\begin{aligned} \dot{\vec{r}} &= \vec{v} \\ \dot{\vec{v}} &= -\frac{GM}{r^2} \hat{r} \end{aligned}}$$

Integrals of Motion

- quantities conserved along trajectories (i.e. in time)

(1) total energy

$$E = K + U = \frac{1}{2} \mu v^2 - \frac{GM\mu}{r}$$

- if $E < 0 \rightarrow$ bound system $\rightarrow$ elliptical orbits
- if $E > 0 \rightarrow$ unbound i.e. scattering $\rightarrow$ hyperbolic 'orbits'

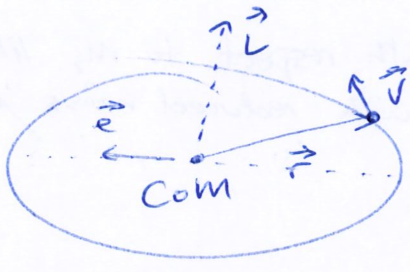
(2) angular momentum

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times \mu \vec{v}$$

specific ang. mom.

$$\vec{j} = \vec{L}/\mu$$

[is perpendicular to the orbit, which is confined to a plane]

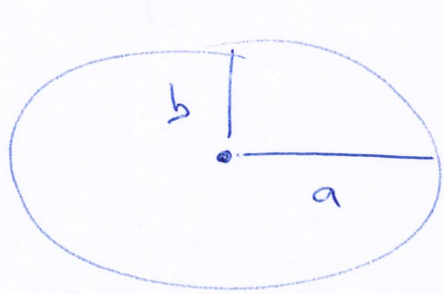


(3) Laplace-Runge-Lenz vector

$$\vec{e} = \frac{\vec{v} \times \vec{j}}{GM} - \frac{\vec{r}}{r}$$

→ conserved in all problems with a central force that interacts two bodies as $1/r^2$ ("Kepler problem")
[potential $\propto 1/r$]

also called the "eccentricity vector" of the elliptical orbit



$$|\vec{e}| = e = (1 - b^2/a^2)^{1/2}$$

[is the ratio of semi-major and semi-minor axes, i.e. $e = 0 \Rightarrow$ circle]

<u>idea</u>: these three quantities are conserved in the mathematical i.e. analytic sense, but when approximating the solution numerically, conservation will not be guaranteed

↳ commonly check e.g. conservation of energy to assess the accuracy, and convergence, of a numerical scheme

- reformulate into dimensionless system of equations:

length: $\vec{s} = \vec{r}/R_0$ $\quad R_0 \sim$ mean separation

velocity: $\vec{\omega} = \vec{v}/v_0$ $\quad$ where $v_0 = \left(\frac{GM}{R_0}\right)^{1/2}$

time: $\tau = t/T_0$ $\quad$ where $T_0 = R_0/v_0 = \left(\frac{R_0^3}{GM}\right)^{1/2}$

$$\Rightarrow \boxed{\begin{aligned} \frac{d\vec{s}}{d\tau} &= \vec{\omega} \\ \frac{d\vec{\omega}}{d\tau} &= -\frac{\vec{s}}{s^3} \end{aligned}}$$

coupled set of 2 first-order ODEs to be solved numerically

How? Must discretize time!

- define $h = \tau_i - \tau_{i-1}$ the stepsize / timestep
- take h constant for simplicity

notation: $\vec{s}_i = \vec{s}(\tau_i)$

$$\Rightarrow \frac{d\vec{s}}{d\tau} = \frac{\Delta\vec{s}}{\Delta\tau} = \frac{\vec{s}_i - \vec{s}_{i-1}}{\tau_i - \tau_{i-1}} = \frac{\vec{s}_i - \vec{s}_{i-1}}{h} + \underbrace{\mathcal{O}(h)}$$

have an error i.e. neglected term that is of order h, i.e. linearly proportional to h

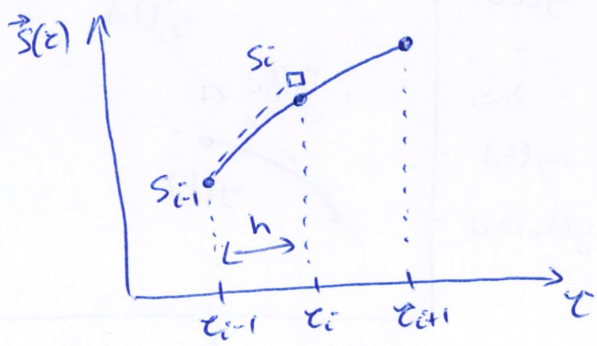
LHS: $\frac{d\vec{s}_i}{d\tau} \rightarrow$ implicit scheme

$\frac{d\vec{s}_{i-1}}{d\tau} \rightarrow$ explicit scheme

$$\frac{d\vec{s}_{i-1}}{d\tau} = \frac{1}{h}(\vec{s}_i - \vec{s}_{i-1}) + \mathcal{O}(h)$$

$$\Rightarrow h\frac{d\vec{s}_{i-1}}{d\tau} = \vec{s}_i - \vec{s}_{i-1} + \mathcal{O}(h^2)$$

$$\Rightarrow \boxed{\vec{s}_i = \vec{s}_{i-1} + h\vec{\omega}_{i-1} + \mathcal{O}(h^2)}$$



and do the same for the 2^{nd} eqn:

$$h\underbrace{\frac{d\vec{\omega}_{i-1}}{d\tau}} = \vec{\omega}_i - \vec{\omega}_{i-1} + \mathcal{O}(h^2) \quad \Rightarrow \boxed{\vec{\omega}_i = \vec{\omega}_{i-1} - h\frac{\vec{s}_{i-1}}{s_{i-1}^3} + \mathcal{O}(h^2)}$$

substitute in $-\vec{s}_{i-1}/s_{i-1}^3$

note: $\{i-1, i\} \rightarrow \{i, i+1\}$ same

Euler's Method

consider a 1^{st} order ODE initial-value problem

$$y' = \frac{dy}{dt} = f(t,y) \qquad a \le t \le b \qquad y(a) = \alpha$$

goal : approximate solution $\tilde{y}(t)$ on a discrete number of 'mesh' or 'grid' points across the interval $[a,b]$

→ Taylor expansion :

$$y(t_{i+1}) = y(t_i) + (t_{i+1} - t_i)\, y'(t_i) + \frac{(t_{i+1}-t_i)^2}{2}\, y''(\xi_i)$$

for some number $\xi_i \in [t_i, t_{i+1}]$

notation : step size $h = t_{i+1} - t_i = \frac{a-b}{N}$ for N points (constant)

$$\Rightarrow \quad y(t_{i+1}) = y(t_i) + h\underbrace{y'(t_i)}_{\text{substitute } y'(t)} + \frac{h^2}{2} y''(\xi_i)$$

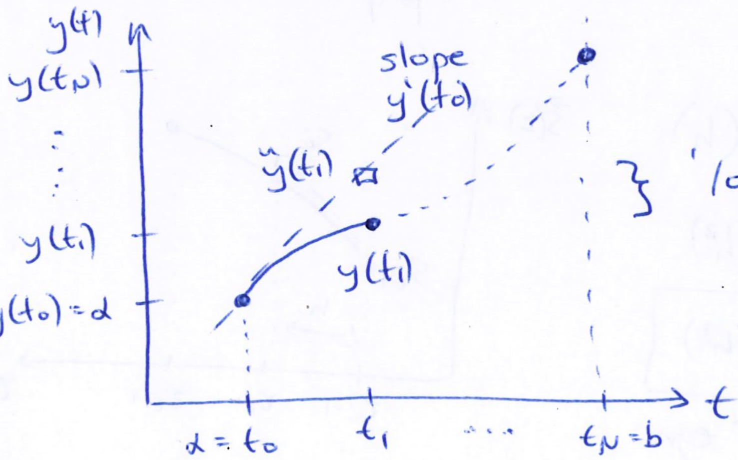
notation: $y(t_i) \equiv y_i$

$$\Rightarrow \quad y_{i+1} = y_i + h f(t_i, y_i) + \frac{h^2}{2} y''(\xi_i)$$

drop remainder term : Euler's method

$$\boxed{\tilde{y}_{i+1} = \tilde{y}_i + h f(t_i, \tilde{y}_i)}$$

geometrically, if $\tilde{y}_i \simeq y_i \Rightarrow f(t_i, \tilde{y}_i) = y_i'$ the slope of $y(t)$ at t_i



} 'local' truncation error

$|\tilde{y}(t_1) - y(t_1)| \sim \mathcal{O}(h^2)$

but ! since total number of steps $N_i = \frac{t_i - t_0}{h} \propto \frac{1}{h}$

⇒ 'global' truncation error

$|\tilde{y}(t_i) - y(t_i)| \sim \mathcal{O}(h)$

- forward Euler ⇒ explicit
- backward Euler ⇒ implicit

$$\boxed{\hat{y}_{i+1} = \hat{y}_i + h f(t_{i+1}, \hat{y}_{i+1})}$$